Circular Motion Worksheet

1. A 0.40 kg ball is attached to a string and moves in a horizontal circle of radius 1.5 m on a frictionless table. The ball completes 2.5 revolutions every second.

(a) Calculate the angular velocity ω of the ball.

(b) Determine the speed of the ball.

(c) Find the tension in the string.

2. A small mass of 0.75 kg is attached to a string and whirled in a horizontal circle of radius 0.80 m. The tension in the string is 48 N.

(a) Calculate the centripetal acceleration of the mass.

(b) Determine the angular velocity ω.

(c) Find the time period of the motion.

3. A car of mass 1200 kg travels around a flat circular track of radius 55 m at a constant speed of 20 m s^{-1}.

(a) Calculate the centripetal acceleration of the car.

(b) Determine the resultant centripetal force required to keep the car moving in a circle.

(c) State which force provides the centripetal force in this situation.

4. A 0.20 kg mass attached to a light string moves in a vertical circle of radius 0.90 m at a constant speed of 5.0 m s^{-1}.

(a) Calculate the required centripetal force.

(b) Determine the tension in the string at the bottom of the circle.

(c) Determine the tension in the string at the top of the circle.

5. A bucket of water of total mass 1.8 kg is swung in a vertical circle of radius 1.2 m. At the top of the circle the tension in the handle is 14 N.

(a) Draw a force diagram for the bucket at the top of the circle.

(b) Calculate the resultant force toward the centre at the top of the circle.

(c) Determine the speed of the bucket.

6. A rollercoaster car of mass 450 kg moves around a vertical circular track of radius 18 m. At the bottom of the loop the normal contact force exerted by the track on the car is 1.25×10^4 N.

(a) Calculate the weight of the car.

(b) Determine the centripetal force acting on the car at the bottom of the loop.

(c) Calculate the speed of the car.

7. A car travels around a banked curve of radius 85 m. The banking angle is 12°. Assume there is no friction.

(a) Draw a force diagram for the car.

(b) Resolve the normal reaction force vertically and horizontally.

(c) Show that the speed required to negotiate the bend without friction is

$$v = \sqrt{rg \tan \theta}$$

(d) Calculate the required speed.

8. A motorcycle travels around a frictionless banked curve of radius 40 m at a speed of 16 m s^{-1}.

(a) Determine the required centripetal acceleration.

(b) Calculate the banking angle needed for the turn.

(c) State why friction is not required at this speed.

9. A 1500 kg car travels around a frictionless banked bend of radius 60 m. The banking angle is 9°.

(a) Calculate the normal reaction force acting on the car.

(b) Determine the horizontal component of the normal force.

(c) Calculate the speed of the car.

Answer Key

1. (a)

$$\omega = 2\pi f = 2\pi(2.5) = 15.7\,\text{rad s}^{-1}$$

(b)

$$v = r\omega = 1.5(15.7) = 23.6\,\text{m s}^{-1}$$

(c)

$$T = \frac{mv^2}{r} = \frac{0.40(23.6)^2}{1.5} \approx 148\,\text{N}$$

2. (a)

$$F_c = ma_c$$

$$a_c = \frac{48}{0.75} = 64\,\text{m s}^{-2}$$

(b)

$$a_c = r\omega^2$$

$$\omega = \sqrt{\frac{64}{0.80}} = 8.94\,\text{rad s}^{-1}$$

(c)

$$T = \frac{2\pi}{\omega} = 0.703\,\text{s}$$

3. (a)

$$a_c = \frac{v^2}{r} = \frac{20^2}{55} = 7.27\,\text{m s}^{-2}$$

(b)

$$F_c = ma_c = (1200)(7.27) \approx 8.73 \times 10^3\,\text{N}$$

(c) Friction between the tyres and the road provides the centripetal force.

4. (a)

$$F_c = \frac{mv^2}{r} = \frac{0.20(5.0)^2}{0.90} = 5.56\,\text{N}$$

(b) Bottom:

$$T - mg = F_c$$

$$T = 5.56 + 1.96 = 7.52\,\text{N}$$

(c) Top:

$$T + mg = F_c$$

$$T = 5.56 - 1.96 = 3.60\,\text{N}$$

5. (a) Both tension and weight act toward the centre.

(b)
$$F_c = T + mg$$
$$F_c = 14 + (1.8)(9.81) = 31.7\,\text{N}$$

(c)
$$\frac{mv^2}{r} = 31.7$$
$$v = \sqrt{\frac{31.7(1.2)}{1.8}} \approx 4.60\,\text{m s}^{-1}$$

6. (a)
$$W = mg = (450)(9.81) = 4410\,\text{N}$$

(b)
$$F_c = R - W$$
$$F_c = 12500 - 4410 = 8090\,\text{N}$$

(c)
$$\frac{mv^2}{r} = 8090$$
$$v = \sqrt{\frac{8090(18)}{450}} \approx 18.0\,\text{m s}^{-1}$$

7. (a) —

(b)
$$R\cos\theta = mg$$
$$R\sin\theta = \frac{mv^2}{r}$$

(c) Dividing equations:
$$\tan\theta = \frac{v^2}{rg}$$

therefore
$$v = \sqrt{rg\tan\theta}$$

(d)
$$v = \sqrt{(85)(9.81)\tan 12^\circ} \approx 13.3\,\text{m s}^{-1}$$

8. (a)
$$a_c = \frac{v^2}{r} = \frac{16^2}{40} = 6.4\,\text{m s}^{-2}$$

(b)

$$\tan\theta = \frac{v^2}{rg}$$

$$\theta = \tan^{-1}\left(\frac{16^2}{40(9.81)}\right) \approx 33.1^\circ$$

(c) The horizontal component of the normal reaction alone provides the required centripetal force.

9. (a)

$$R\cos 9^\circ = mg$$

$$R = \frac{1500(9.81)}{\cos 9^\circ} \approx 1.49 \times 10^4\,\text{N}$$

(b)

$$R\sin 9^\circ \approx 2330\,\text{N}$$

(c)

$$\frac{mv^2}{r} = 2330$$

$$v = \sqrt{\frac{2330(60)}{1500}} \approx 9.66\,\text{m s}^{-1}$$